Efficient Algorithm for Large Scale Electric Vehicle Routing Problem With Charging Capacity Constraint

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***Abstract*— This paper proposes a very efficient algorithm for large scale Electric Vehicle Routing Problem (EVRP) with charging capacity constraint of charging facilities which has not been properly addressed yet. EVRP is to jointly optimize the sequence of serving a set of transportation tasks and the charging schedule for a fleet of electric vehicles (EVs) under the constraints on the State of Charge (SOC) of battery. EVRP has gained intensive attention in recent years, yet, it is a typical mixed integer programming (MIP), which is NP-hard thus intractable for large scale cases. Meanwhile, the maximal power capacity constraint of charging facilities is very important in practice, but has not been properly addressed in previous studies, because it will further complicate the computation. This paper proposes an efficient algorithm to solve large scale EVRP with power capacity constraint of charging stations. The algorithm approximates the original problem with three sequential subproblems that are all linear programming (LP), thus ensuring certain local optimality. Extensive simulations validate the efficiency of the proposed method and show that it is hundreds of times faster than solver and can solve large scale EVRP instances with 1000 tasks.**

***Index Terms*— Electric vehicle, routing problem, mixed integer programming, linear programming.**

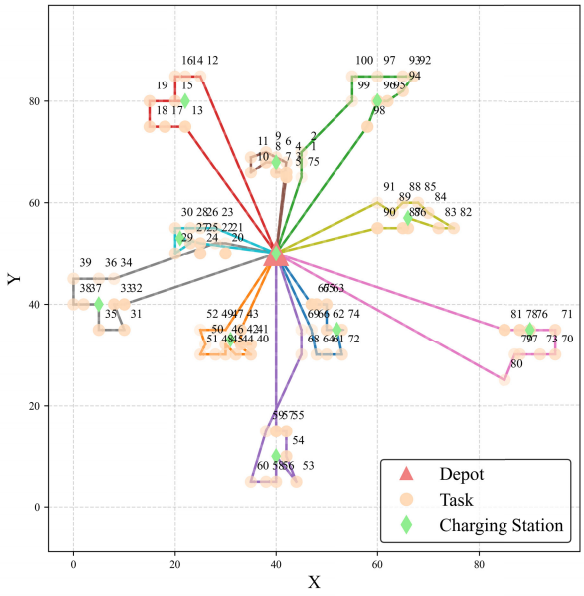


Fig. 1. A schematic diagram illustrating the Electric Vehicle Routing Problem (EVRP). The diagram showcases a scenario involving 10 electric vehicles, 100 transportation tasks, and 11 charging stations. Solid lines represent the planned routes for the vehicles, with each color indicating the route of a specific vehicle.

I. Introduction

THE ever-increasing needs of sustainable transportation have led to a huge and profound change with the vigorous development of the electric vehicle (EV) industry in the past decade [1], [2]. Many transportation companies have replaced the traditional internal combustion engine vehicles to EVs due to the environmental friendliness and low operation cost. These companies often manage a fleet of EVs to handle a variety of geographically dispersed tasks, including city logistics such as parcel delivery, mail collection and distribution, freight transportation, and even passenger mobility services in urban environments. Meanwhile, charging during the trips also needs to be considered due to the limited battery capacity of EVs [3]. This necessity is exemplified in Fig. 1, where the routes of 10 EVs must be determined to fulfill 100 tasks while considering recharging to replenish the battery.

Mathematically, such an operational problem is characterized as the Electric Vehicle Routing Problem (EVRP), which aims at optimizing the service reward and charging cost while satisfying the state of charge (SOC) constraints of EVs, by properly selecting the transportation routes and scheduling the charging. Indeed, EVRP is an important variant of vehicle routing problem (VRP), which is one of the most widely studied problems in the field of operations research. The main technical challenge of EVRP lies at its nature of NP-hardness [4], because it is a typical mixed-integer

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programming (MIP) that contains both integer variable (routes) and continuous variable (charging power).

A. Related Works

The EVRP has become an important research topic due to the growing adoption of EVs in logistics and transportation [5], [6], [7], [8], [9], [10], [11], [12]. To address the EVRP, various solution methodologies have been proposed in the literature. These approaches could be broadly divided into three classes: analytic, heuristic and learning-based approaches. Analytic methods for EVRP mainly involve mathematical modeling and exact algorithms, such as branch and cut [13], branch-price-and-cut [14], branch-and-cut-and-price [15] and branch-and-price [16] etc. These approaches provide accurate solutions but are computationally intensive, making them only suitable for smaller problem instances. However, these methods are not suitable for real-world scenarios with a larger number of customers. For example, Lee [16] can only solve instances with 36 customers in 1-hour time limit.

Heuristic approaches focus on developing practical algorithms to efficiently solve EVRP. These encompass construction heuristics, meta-heuristics and matheuristics. For the construction heuristics, the most common treatment is to remove or insert charging stations for a route, which has been applied in several studies, such as [4], [5], [17], and [18]. These algorithms have been tested on instances with 100 customers. Meanwhile meta-heuristic algorithms are widely used to solve EVRP, such as large neighborhood search/adaptive large neighborhood search [4], [19], [20], variable neighborhood search/adaptive variable neighborhood search [21], [22], [23], tabu search/adaptive tabu search/granular tabu search [24], [25], [26]. Apart from these methods, there are matheuristic methods that combine analytic and heuristic methods to enhance performance [13], [27]. These heuristics can achieve near-optimal solutions within much less time than analytical methods. However, these methods still have some limitations, e.g., they are typically applied to scenarios with fewer than 200 customers, and the required computation time can still be lengthy. For instance, in [25], the customer limit is set at 100, with an average computation time of 412.82 seconds.

Beyond analytic methods and heuristic methods, there is a growing trend in the use of learning-based approaches, such as [28] and [29]. While learning-based optimization methods hold great potential for addressing this problem, there is a notable scarcity of research in this area. For those who are more interested in EVRP, we recommend referring to [30], which provides a literature review on this problem.

Overall, the existing methods still suffer from slow computation speeds, particularly when dealing with large-scale problems, making them impractical for real-world applications. Therefore, the first motivation of this research is to develop new methodology that is able to achieve near-optimal solutions with fast speed for large scale problems. The second motivation is that the charging capacity constraints of charging stations are often insufficiently addressed in the majority of existing research. A few recent works, e.g., [31], [32], [33] consider the power capacity of charing stations. However, their approaches are still based on MIP algrithms and solvers, thus are computation intensive and only suitable for small-scale problems. For example, [31] is the most related work to ours, which also considers the power capacity constraint of charging stations in their mathematical model, but uses solver for a solution. As a result, their approach can only handle small scale problems. Specifically, the number of customers was limited to a maximum of 15, with no more than 5 EVs, and the total number of charging visits for all vehicles to the station did not exceed 3. The experimental settings and results differ significantly from real-world scenarios, and therefore fail to fully demonstrate the true impact of the power capacity constraint of charging stations.

B. Contribution

In summary, current research has not satisfactorily addressed the challenges of large scale and charging capacity constraints for EVRP. In this paper, we propose a highly efficient algorithm based on linear programming designed to address such challenges. The key contributions of this study are outlined below:

1) We fully address the EVRP with charging capacity constraints, which distinguishes our work from traditional EVRP studies.
2) Mathematically, we formulate a unified graph model to characterize both serving and charging actions of EVs. In this model, EV charging behavior is represented by visits to a set of charging nodes.
3) The original EVRP is approximated by three sequential subproblems, i.e., vehicle routing, determining charging stations, and scheduling charging power. Importantly, all three subproblems are structured as LPs, ensuring efficient solutions and local optimality.
4) Extensive numerical experiments, complemented by comprehensive comparisons with solver and adaptive large neighborhood search, are conducted to validate the effectiveness and efficiency of the proposed method.

The remainder of the paper is structured as follows. In Section II, we present the system model and outline the problem formulation. The proposed algorithm for addressing the large scale EVRP with charging capacity constraints is thoroughly discussed in Section III. Section IV is dedicated to presenting numerical results. Finally, Section V provides a conclusion to this study.

II. System Model and Problem Formulation

In this section, we begin by introducing a graph encompassing key elements, including EVs, transportation tasks, and charging stations. Subsequently, we formulate the EVRP as an optimization problem on this graph. To enhance clarity and facilitate a better understanding of our model, we provide Table I, offering a concise reference to the key notations employed in our formulation.

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TABLE I
KEY NOTATIONS

Decision Variables	
x_{ij}^k	Binary variable indicating whether EV k traverses edge e_{ij}
p_i^k	Continuous variable, charging power of EV k at node i
y_j^k	Auxiliary decision variable of SOC constraint
Parameters	
t_i	Arrival/Start time of node i
T_{ij}	Traveling time from node i to j
d_{ij}	Distance from node i to j
w_{ij}^k	Energy consumption percentage of EV k from node i to j
r_i	Service reward of node i
q_i	Electricity price of charging node i
ΔT	Time duration of the charging nodes
BC^k	Battery capacity of EV k
$\bar{p}_i$	Maximum charging power of an EV
P_i	Maximum total charging power of a charging station
Other Notations	
$\mathcal{G}(\mathcal{V},\mathcal{E})$	The directed acyclic graph
$\mathcal{V}$	Nodes of the graph
$\mathcal{E}$	Edges of the graph
$\mathcal{T}$	Set of the transportation task nodes
$\mathcal{C}$	Set of the charging nodes
$\mathcal{K}$	Set of electric vehicles
Z_T^k	Route of EV k only with task nodes
Z_C^k	Route of EV k only with charging nodes
$Z_{T,C}^k$	Route of EV k with task nodes and charging nodes
s_i^k	SOC when EV k arrives at node i

A. Graph Model

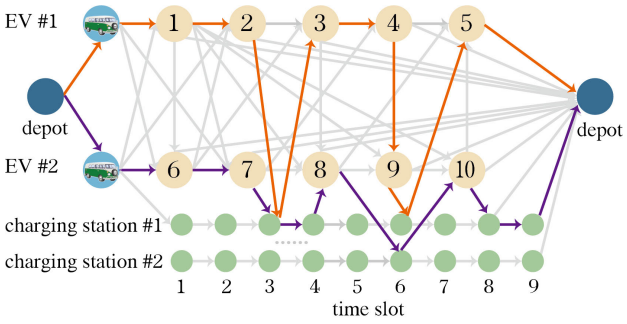
We consider a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{v_o, v_d\} \cup \mathcal{T} \cup \mathcal{C}$ is the union of the origin and destination depots $\{v_o, v_d\}$, transportation task nodes $\mathcal{T}$, and the charging nodes $\mathcal{C}$. $\mathcal{E}$ is the set of edges that connect the nodes.

- $\{v_o, v_d\}$ represents a pair of depots that a fleet of homogeneous EVs $\mathcal{K}$ start from and end at.
- For a transportation task node $i \in \mathcal{T}$, it is characterized by a tuple $\{l_i, t_i, r_i\}$ that specifies the geographical location, arrival time and service reward of the task. To accomplish this task, an EV must arrive at l_i before t_i.
- For a charging node $i \in \mathcal{C}$, it is also characterized by a similar tuple $\{l_i, t_i, q_i\}$, which stands for a charging action in a charging station l_i starting from time slot t_i for a constant period ΔT at the electricity price q_i. It is noteworthy that employing varying values for q_i allows for the representation of fluctuations in electricity prices across different time slots and locations.

It is notable that with this graph model, different time slots for charging at the same charging station correspond to distinct charging nodes. To illustrate, a charging station maps to a series of charging nodes, as depicted in Fig. 2. Furthermore, in this model, the charging power of an EV at a charging station is not an attribute of the charging node but rather a decision variable to be determined later.

B. Optimization Model of EVRP

1) Objective Function: Let $x_{ij}^k \in \{0, 1\}$ be the binary decision variable denoting whether an EV $k \in \mathcal{K}$ traverses an edge $e_{ij} \in \mathcal{E}$ that is directed from node $i \in \mathcal{V}$ to node $j \in \mathcal{V}$. Meanwhile, let p_i^k be the continuous decision variable standing for the charging power of EV k at node i. With no doubt, the charging power p_i^k is only meaningful when i is a charging node, i.e., $i \in \mathcal{C}$.



(Partial edges of the EVRP Graph are omitted)

Fig. 2. The EVRP graph, with a single pair of origin and destination depots, 2 charging stations (#1 and #2), 2 EVs (#1 and #2) and 10 customers. The two depot nodes are v_o and v_d, nodes with EV are virtual vehicle nodes, the brown nodes and green nodes are task nodes and charging nodes, respectively. Some edges of the graph are omitted due to simplification.

The goal is to enhance the service reward while minimizing the traveling cost and charging cost. Consequently, the objective function can be expressed as,

$$\min_{x_{ij}^k, p_i^k} \sum_{k\in\mathcal{K}} (\omega_1 \sum_{ij\in\mathcal{E}} d_{ij} x_{ij}^k - \omega_2 \sum_{j\in\mathcal{T}} r_j x_{ij}^k + \omega_3 \sum_{i\in\mathcal{C}} q_i p_i^k \Delta T x_{ij}^k), \tag{1}$$

where ω_1, ω_2 and ω_3 are coefficients of traveling distance, service reward and charging cost respectively. Next, we shall present the constraints of EVRP.

2) Network Flow Constraints: As the EVRP is constructed using a graph model, adherence to the following standard network flow constraints is imperative. Constraint (2) represents the flow conservation constraint, ensuring a balance between input flow and output flow.

$$\sum_{i\in\mathcal{V}} x_{ij}^k - \sum_{i\in\mathcal{V}} x_{ji}^k = b_j, \forall j \in \mathcal{V}, k \in \mathcal{K},$$
$$b_j = \begin{cases} -1 & j = v_o \\ 1 & j = v_d \\ 0 & j \in \mathcal{T} \cup \mathcal{C}. \end{cases} \tag{2}$$

Constraint (3) requires that the task node could not be visited more than once as below,

$$\sum_{k\in\mathcal{K}} \sum_{i\in\mathcal{V}} x_{ij}^k \le 1, \forall j \in \mathcal{T}. \tag{3}$$

It is important to note that this constraint does not apply to charging nodes $j \in \mathcal{C}$, as multiple EVs can visit the same charging station. Constraint (4) prohibits a node from connecting to itself, thus preventing self-loops,

$$x_{i,i}^k = 0, \forall i \in \mathcal{V}, k \in \mathcal{K}. \tag{4}$$

Additionally, time compatibility is guaranteed by constraint (5),

$$t_j \ge (t_i + T_{ij}) x_{ij}^k, \qquad \forall i, j \in \mathcal{V}, k \in \mathcal{K}. \tag{5}$$

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It should be noted that T_{ij} is the known average traveling time between node i and node j. However, if node i is a charging node, the charging time duration ΔT is also included in T_{ij}.

3) SOC and Charging Constraints: With the traveling and charging of EV, its SOC will also change over time. Let s_j^k be the SOC when an EV k arrives at a node j, thus its dynamics can be readily represented as below,

$$s_j^k = \sum_{i\in\mathcal{C}}(s_i^k + \frac{p_i^k \Delta T}{BC^k} - w_{ij}^k)x_{ij}^k + \sum_{i\in\mathcal{V}\setminus\mathcal{C}}(s_i^k - w_{ij}^k)x_{ij}^k, \forall j \in \mathcal{V}, k \in \mathcal{K}. \quad (6)$$

where w_{ij}^k is the energy consumption percentage from i to j. Clearly, SOC must be operated within a certain range, and the battery of EV should be prevented from overcharging.

$$\underline{s}\sum_{i\in\mathcal{V}} x_{ij}^k \le s_j^k \le \bar{s}, \forall j \in \mathcal{V}, k \in \mathcal{K}, \quad (7a)$$

$$s_j^k + \frac{p_j^k \Delta T}{BC^k} \le \bar{s}, \forall j \in \mathcal{C}, k \in \mathcal{K}. \quad (7b)$$

Meanwhile, the individual charging power constraint for each EV,

$$0 \le p_i^k \le \bar{p}_i, \forall i \in \mathcal{C}, k \in \mathcal{K}, \quad (8)$$

and total charging power constraint for the entire station,

$$0 \le \sum_{k\in\mathcal{K}} p_i^k \le \bar{P}_i, \forall i \in \mathcal{C}, \quad (9)$$

must be satisfied in the charging process. The constants $\bar{p}_i$ and $\bar{P}_i$ stand for the maximal power can be sent to a single EV, and the maximal total charging power of a charging station.

4) Optimization Model: Consequently, the complete EVRP is formulated as the following optimization model on the graph $\mathcal{G}$,

Original EVRP:

$$\min_{x_{ij}^k, p_i^k} \sum_{k\in\mathcal{K}}(\omega_1 \sum_{ij\in\mathcal{E}} d_{ij}x_{ij}^k - \omega_2 \sum_{j\in\mathcal{T}} r_j x_{ij}^k + \omega_3 \sum_{i\in\mathcal{C}} q_i p_i^k \Delta T x_{ij}^k)$$
$$s.t. \quad (2)\text{-}(9),$$
$$x_{ij}^k \in \{0,1\}, \forall i, j \in \mathcal{V}, k \in \mathcal{K},$$
$$p_i^k \ge 0, \forall i \in \mathcal{C}, k \in \mathcal{K}. \quad (10)$$

Clearly, above problem is a typical MIP because it contains both integer variables x_{ij}^k and continuous variables p_i^k. Due to its NP-hardness, it is computational expensive and time-consuming, and intractable especially for large scale applications. In addition, besides (3), the charging capacity constraint (9) also couples the decisions across different EVs, which further complicates the computation. To this end, a novel and efficient matheuristic algorithm is detailed next to tackle this problem.

III. Methodology

The core idea of our proposed method revolves around approximating the primary problem with three sequential subproblems, i.e. vehicle routing, inserting charging nodes, and charging scheduling. Each subproblem is an LP in nature, thus ensuring computational efficiency and local optimality. The algorithm is detailed below.

A. Vehicle Routing

In real-world applications, serving more customers is of a higher priority for EVRP, thus we consider the task routing first. If we initially disregard the SOC and charging constraints (6)-(9), the remaining subproblem shown below is a conventional vehicle routing problem (VRP) on the acyclic graph $\mathcal{G}$.

Vehicle Routing Subproblem:

$$\min_{x_{ij}^k} \sum_{k\in\mathcal{K}}(\omega_1 \sum_{ij\in\mathcal{E}} d_{ij}x_{ij}^k - \omega_2 \sum_{j\in\mathcal{T}} r_j x_{ij}^k),$$
$$s.t. \quad (2)\text{-}(5),$$
$$x_{ij}^k \in \{0,1\}, \forall i, j \in \mathcal{V}, k \in \mathcal{K}. \quad (11)$$

It is notable that the charging cost term $q_i p_i^k$ is omitted here, because no EV will go to charging nodes that in fact reduce the overall revenue, if SOC constraints are not considered. For the same reason, the routing only needs to be performed on a subgraph with task nodes and depots. Clearly, the routing subproblem is a pure ILP instead of MIP, and only determines the integer variables x_{ij}^k.

Proposition 1: The vehicle routing subproblem (11) could be converted into an equivalent LP form (18).

Proof: Since all EVs are assumed to be homogeneous, the 3-index variable x_{ij}^k can be replaced by the 2-index variable $x_{ij} \in \{0,1\}$ without changing the solution. With the replacement, the objective in (11) becomes

$$\min_{x_{ij}} \omega_1 \sum_{ij\in\mathcal{E}} d_{ij}x_{ij} - \omega_2 \sum_{j\in\mathcal{T}} r_j x_{ij}. \quad (12)$$

Similarly, the constraints (2)-(5) will be modified accordingly,

$$\sum_{i\in\mathcal{V}} x_{ij} - \sum_{i\in\mathcal{V}} x_{ji} = b_j, \forall j \in \mathcal{V},$$
$$b_j = \begin{cases} -|\mathcal{K}| & j = v_o \\ |\mathcal{K}| & j = v_d \\ 0 & j \in \mathcal{T}. \end{cases} \quad (13)$$
$$\sum_{i\in\mathcal{V}} x_{ij} \le 1, \forall j \in \mathcal{T}, \quad (14)$$
$$x_{i,i} = 0, \forall i \in \mathcal{V}, \quad (15)$$

and

$$t_j \ge (t_i + T_{ij})x_{ij}, \qquad \forall i, j \in \mathcal{T} \cup \{v_o, v_d\}. \quad (16)$$

Clearly, the objective (12) is linear in x_{ij}, and the constraints (13) and (14) satisfy the unimodular condition [34]. So the coefficient matrix of (13) and (14) is totally unimodular (TU). In addition, constraint (15) could be seen as a bound constraint. Constraint (16) could be transformed to

$$x_{ij} \le \lfloor \frac{t_j}{t_i + T_{ij}} \rfloor, \forall i, j \in \mathcal{V}, \quad (17)$$

which also serves as a bound constraint. Hence the coefficient matrix of the constraint (15) and (17) are part of identity

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matrix $I^{|\mathcal{V}|\times|\mathcal{V}|}$. According to Proposition 3.1 in [34], the matrix (A, I) is TU if matrix A is TU. So it can be proven that the coefficient matrix of (13)-(15) and (17) is TU. As a result, we can exactly relax the integer variable $x_{ij} \in \{0, 1\}$ to continuous variable $x_{ij} \in [0, 1]$, thus convert the ILP problem to the following LP problem,

LP Routing Subproblem

$$\begin{aligned}&\min_{x_{ij}} \omega_1 \sum_{ij\in\mathcal{E}} d_{ij}x_{ij} - \omega_2 \sum_{j\in\mathcal{T}} r_j x_{ij},\\ &s.t.\quad \text{(13)-(15), (17)},\\ &\qquad x_{ij} \in [0,1], \forall i, j \in \mathcal{V}.\end{aligned} \tag{18}$$

□

In this manner, an expeditious routing solution for the EV fleet can be obtained. Essentially, each EV is assigned a route that intricately links a sequence of tasks. Subsequently, we embark on the insertion of charging nodes into these routes, with the concurrent determination of the optimal charging power levels for effective fleet management.

B. Inserting Charging Nodes

To charge the battery, an EV shall first have access to charging stations and then determine the charging power. Therefore, in order to increase the opportunity of accessing to charging stations, in this step, we aim to insert as many charging nodes as possible into the routes obtained in the previous subproblem.

Consider the route of EV k, which is a sequence of task nodes and depots $Z_T^k = (j_1, \ldots, j_m, j_{m+1}, \ldots), j \in \mathcal{T} \cup \{v_o, v_d\}$ that shall be traversed one-by-one by EV k. Clearly, we shall insert as many charging nodes as possible to the intervals between any two consecutive nodes in the path, for example, j_m and j_{m+1}. Mathematically, it can be written as an optimization problem shown below,

Charging Node Inserting Subproblem:

$$\begin{aligned}&\min_{x_{ij}} \omega_1 \sum_{i,j\in Z_T^k\cup\mathcal{C}} d_{ij}x_{ij} - \omega_4 \sum_{i,j\in Z_T^k\cup\mathcal{C}} r_j x_{ij}\\ &s.t.\ \sum_{i\in\mathcal{V}} x_{ij} - \sum_{i\in\mathcal{V}} x_{ji} = b_j, \forall j \in Z_T^k \cup \mathcal{C},\\ &\qquad b_j = \begin{cases} -1 & j = v_o\\ 1 & j = v_d\\ 0 & j \in \mathcal{T}\cup\mathcal{C}.\end{cases}\\ &\qquad t_j \geq (t_i + T_{ij})x_{ij}, \forall i, j \in Z_T^k \cup \mathcal{C},\\ &\qquad x_{ij} \in \{0,1\}, \forall i, j \in Z_T^k \cup \mathcal{C}.\end{aligned} \tag{19}$$

In the objective function of this subproblem, apart from the traveling cost, we also consider a penalty term consisting of reward. The setting of the reward is to ensure that the task nodes and charging nodes are incorporated in the route. However, because we use charging nodes to model the charging station, considering all of the charging nodes can significantly increase the number of variables and constraints, thereby slowing down the algorithm. To address this issue, we adopt a preprocessing operation here. We limit the range of the charging nodes according to spatial criteria. Given j_m and j_{m+1}, we could obtain the nearest charging station through greedy method. Hence the charging nodes set $\mathcal{C}$ could be replaced by the selected one $\mathcal{C}'$. Similar to previous analysis, above ILP problem (19) also satisfies the totally unimodular condition thus can be exactly relax to an LP problem by letting $x_{ij} \in [0, 1]$, as shown in (20).

LP Charging Node Inserting Subproblem:

$$\begin{aligned}&\min_{x_{ij}} \omega_1 \sum_{i,j\in Z_T^k\cup\mathcal{C}'} d_{ij}x_{ij} - \omega_4 \sum_{i,j\in Z_T^k\cup\mathcal{C}'} r_j x_{ij}\\ &s.t.\ \sum_{i\in\mathcal{V}} x_{ij} - \sum_{i\in\mathcal{V}} x_{ji} = b_j, \forall j \in Z_T^k \cup \mathcal{C}',\\ &\qquad b_j = \begin{cases} -1 & j = v_o\\ 1 & j = v_d\\ 0 & j \in \mathcal{T}\cup\mathcal{C}'\end{cases}\\ &\qquad x_{ij} \leq \lfloor \frac{t_j}{t_i + T_{ij}} \rfloor, \forall i, j \in Z_T^k \cup \mathcal{C}',\\ &\qquad x_{ij} \in [0,1], \forall i, j \in Z_T^k \cup \mathcal{C}'.\end{aligned} \tag{20}$$

Meanwhile, it is notable that the inserting subproblem can be independently solved by each EV, thus enabling parallel computation. After computing, the routes including charging nodes, $Z_{T,C}^k$, could be obtained. Subsequently, the charging power will be determined based on these routes.

C. Charging Scheduling

Based on the route $Z_{T,C}^k$, we consider the route of EV k only with charging nodes, Z_C^k. Considering the SOC dynamics of EV k along its route Z_C^k, the energy consumption $w_{n,n+1}$ between any two consecutive charging nodes j_n and j_{n+1} is known. If there are task nodes between j_n and j_{n+1}, $w_{n,n+1}$ equals the accumulated energy consumption of serving these tasks; otherwise, $w_{n,n+1} = 0$ if no task node is in between them. Consequently, the SOC dynamics of EV k along Z_C^k is given below,

$$s_{j_{n+1}}^k = s_{j_n}^k + \frac{p_{j_n}^k \Delta T}{BC^k} - w_{n,n+1}. \tag{21}$$

Certainly, in addition to ensuring the fulfillment of the box constraints on SOC and charging power (7)-(9), it is imperative to consider the objective of minimizing the charging cost. Thus, the charging scheduling subproblem can be formulated as follows,

LP Charging Scheduling Subproblem:

$$\begin{aligned}&\min_{p_j^k} \sum_{k\in\mathcal{K}} \sum_{j\in Z_C^k} q_j p_j^k\\ &s.t.\ \text{(21)}, \forall k \in \mathcal{K},\\ &\qquad \underline{s} \leq s_j^k \leq \bar{s}, \forall j \in Z_C^k, \forall k \in \mathcal{K},\\ &\qquad s_j^k + \frac{p_j^k \Delta T}{BC^k} \leq \bar{s}, \forall j \in Z_C^k, \forall k \in \mathcal{K},\\ &\qquad 0 \leq p_j^k \leq \bar{p}_j, \forall j \in Z_C^k, \forall k \in \mathcal{K},\\ &\qquad 0 \leq \sum_{k\in\mathcal{K}} p_j^k \leq \bar{P}_j, \forall j \in \cup_{k\in\mathcal{K}} Z_C^k.\end{aligned} \tag{22}$$

Clearly, above subproblem again is an LP and readily solvable.

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However, it is notable that (22) may be infeasible if insufficient number of charging nodes are inserted into the path when the tasks are too tight in time. To deal with this issue, we can relax the box constraint of SOC, $\underline{s} \le s_j^k \le \bar{s}$, and revise (22) as below,

Relaxed LP Charging Scheduling Subproblem:

$$\begin{aligned}
&\min_{p_j^k, y_j^k} \sum_{k\in\mathcal{K}} \sum_{j\in Z_C^k} (q_j p_j^k + \alpha y_j^k) \\
&s.t.\ (21), \forall k \in \mathcal{K}, \\
&\quad \underline{s} - s_j^k \le y_j^k, \forall j \in Z_C^k, \forall k \in \mathcal{K}, \\
&\quad s_j^k - \bar{s} \le y_j^k, \forall j \in Z_C^k, \forall k \in \mathcal{K}, \\
&\quad s_j^k + \frac{p_j^k \Delta T}{BC^k} \le \bar{s}, \forall j \in Z_C^k, \forall k \in \mathcal{K}, \\
&\quad 0 \le y_j^k, \forall j \in Z_C^k, \forall k \in \mathcal{K}, \\
&\quad 0 \le p_j^k \le \bar{p}_j, \forall j \in Z_C^k, \forall k \in \mathcal{K}, \\
&\quad 0 \le \sum_{k\in\mathcal{K}} p_j^k \le \bar{P}_j, \forall j \in \cup_{k\in\mathcal{K}} Z_C^k. \qquad (23)
\end{aligned}$$

where $\alpha > 0$ is a large penalty weight, and y_j^k is an auxiliary decision variable that quantifies how much the SOC box constraint is violated. If no violation occurs, $y_j^k = 0$ and the relaxed subproblem (23) is identical to (22); otherwise, $y_j^k > 0$ implies the SOC box constraint is violated, an iterative modification process is proposed next to tackle this issue.

D. Iterative Modification

Since the relaxed subproblem (23) is always feasible, we can yield the value of y_j^k. Let j^- be the first node in path $Z_{T,C}^k$, such that $y_{j^-}^k > 0$. It implies the SOC box constraint of EV k is violated for the first time at node j^-, i.e., $s_{j^-}^k < \underline{s}$ or $s_{j^-}^k > \bar{s}$.

Then, we can remove the least beneficial task node prior to node j^- from the path $Z_{T,C}^k$ and recompute the charging node inserting subproblem (20). Here, the benefit of serving a node j can be defined as $\frac{1}{\Delta d_j}$, and Δd_j is the extra energy needed for serving node j instead of bypassing it. Conceptually, an EV prefers charging over serving the task node with low benefit. With more charging nodes contained in path $Z_{T,C}^k$, we can reschedule the charging power according to (23). The "remove-insert-reschedule" cycle will be executed until no SOC violation occurs for all EVs.

Finally, the overall computing process is summarized as Algorithm 1. A flowchart is given in Fig.3 to summarize the whole procedure of the proposed method.

E. Complexity Analysis

Since the vehicle routing subproblem (18), charging node inserting subproblem (20) and charging scheduling subproblem (23) are all LP problems, we consider the total number of LP problems solved in the Algorithm 1. It should be noted that, we take a short cut when inserting charging nodes into the route to accelerate the algorithm. If there is no task in the route Z_T^k or all y_j^k are 0, the inserting subproblem will be skipped for EV k.

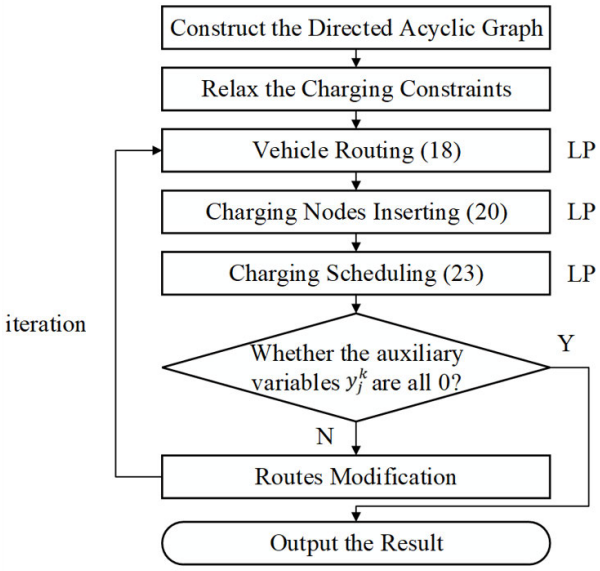


Fig. 3. The flowchart of Algorithm 1.

Algorithm 1 EVRP-3LP

Input: $\mathcal{T}$, $\mathcal{K}$
Output: Z_{final}^k, P_{final}^k, S_{final}^k, $\forall k \in \mathcal{K}$

```
flag = False
Solve vehicle routing subproblem (18)
while flag = False do
    for k ∈ K do
        Solve Problem (20) to insert charging nodes
    end for
    Solve Problem (23) to schedule the charging power
    if all y_j^k = 0 then
        flag = True
    else
        Modify the route of EV k if there is y_j^k ≠ 0
    end if
end while
```

Proposition 2: Let n_{LP} be the total number of LPs solved in the Algorithm 1, then $3 \le n_{LP} \le 2|\mathcal{T}| + |\mathcal{K}| + 2$.

Proof: We first consider the lower bound of n_{LP}. We assume that the algorithm ends in the 0-th iteration, all the tasks are assigned to only one EV, and all y_j^k are 0, then only one vehicle routing subproblem, one inserting subproblem and one charging scheduling problem are solved, hence the lower bound of n_{LP} is 3. As for the upper bound of n_{LP}, we consider the algorithm iterations. In the 0-th iteration, at most $|K| + 2$ LPs are computed. In the i-th ($i \ge 0$) iteration, if $n_{del}^i (n_{del}^i \ge 1)$ tasks are deleted from the route, then n_{del}^i inserting problems and one charging scheduling problem will be computed in the next iteration. Then we have $n_{del}^i + 1 \le 2n_{del}^i$, since $\sum_i n_{del}^i \le |\mathcal{T}|$, then $\sum_i (n_{del}^i + 1) \le \sum_i 2n_{del}^i = 2\sum_i n_{del}^i \le 2|\mathcal{T}|$. Hence $n_{LP} \le 2|\mathcal{T}| + |\mathcal{K}| + 2$. □

It should be pointed out that $2|\mathcal{T}|+|\mathcal{K}|+2$ is a very relaxed upper bound, which will be revealed in Section IV.

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TABLE II
DATASET FOR EXPERIMENT

Dataset	No.	Instance	Customers	EVs	CSs
Solomon	1	c101	100	10	11
	2	c201	100	10	11
	3	r101	100	25	36
	4	r201	100	25	50
	5	rc101	100	25	10
	6	rc201	100	25	20
Errico	7	c101_21_100	100	10	21
	8	c201_21_100	100	10	21
	9	r101_21_100	100	25	21
	10	r201_21_100	100	25	21
	11	rc101_21_100	100	25	21
	12	rc201_21_100	100	25	21
Homberger	13	c1_8_1	800	50	21
	14	c2_8_1	800	50	22
	15	r1_8_1	800	50	289
	16	r2_8_1	800	50	27
	17	rc1_8_1	800	50	289
	18	rc2_8_1	800	50	18
Belgium	19	n50_k10	50	10	4
	20	n1000_k20	1000	25	10

IV. NUMERICAL STUDY

The proposed approach is tested on datasets with different problem scales from different sources. All the experiments are carried out on a personal computer with Intel Core i7-8750H CPU clocking at 2.2 GHz and 16 GB of RAM. The model and the algorithm are implemented with Python. The linear programming problems within the algorithms are solved with the commercial solver Gurobi 11.0.1. The code repository can be found at https://github.com/zjumhy97/evrp.

A. Experiment Settings

1) Dataset: Our experimental instances are derived from four widely used datasets: Solomon's VRPTW dataset [35], Errico's EVRPTW dataset [7], Homberger's VRPTW dataset [36] and the Belgium VRP dataset [35]. The first three datasets are based on generated two-dimensional maps, while the Belgium dataset utilizes a real-world road network map of Belgium. Notably, only Errico's dataset is specifically designed for the EVRP, whereas the other datasets are originally VRP datasets that have been adapted for the EVRP context. It is important to highlight that our approach focuses on precise customer time points rather than broader time windows. Consequently, the instances we employ in our experiments feature very narrow customer time windows, with customer's time point set as the midpoint of the time window. The specific information of the instances is shown in Table II.

2) Parameter Settings: We adopt the homogeneous EVs whose battery capacity is 60kWh, and the maximum of the charging power is set as 30kW [37]. The average velocity of the EVs is assumed to be 40km/h and the energy consumption rate is specified as 0.2 kWh/km. The initial SOC of the EVs is randomly determined in the reasonable range to simulate the real scenario. The charging capacity of the charging station is set as 300 kW. The service reward of the task is set as 150, and the reward of the charging node is set as 10.

When considering electricity pricing, we use Shenzhen's commercial and industrial time-of-use rates. The rates are structured as follows: from 0:00 to 24:00, off-peak rates are applied at 0.3142 CNY per kWh. During the peak hours from 10:00 to 12:00 and 14:00 to 19:00, the rate is 1.5445 CNY per kWh. For the remaining hours, the flat rate is 1 CNY per kWh.

3) Algorithm Hyperparameter Settings: For the duration of the charging nodes, ΔT, we consider two options: for instances with longer times, the duration is set to 60 minutes, while for those with shorter times, it is set to 10 minutes. For the coefficients of the objective function, ω_1, ω_3 and ω_4 are set as 1, 1 and 1, respectively. ω_2 is set according to the map size.

4) Experiment Design: To validate the performance of our proposed method, we conduct comparative experiments using two different approaches. The first approach involves directly solving the original problem (10) using the commercial solver Gurobi. The second approach employs the adaptive large neighborhood search (ALNS) algorithm described in the paper by Keskin and Çatay [4]. We compare the performance of three methods from two dimensions: solution quality and computation time. Then, we apply our method to real-world scenarios to demonstrate its reliability.

B. Solution Quality

1) Optimality Gap: Since Gurobi can only solve small-scale problems and encounters memory limitations when addressing larger problems, we use Gurobi for direct solving only on the Solomon and Errico datasets. We then compare the results to evaluate the solution quality. The optimality gap is calculated using the formula $\Delta = |\frac{\text{Obj}_{3\text{LP}} - \text{Obj}_{\text{Gurobi}}}{\text{Obj}_{\text{Gurobi}}}| \times 100\%$. Figure 4 illustrates the optimality gaps for the two datasets, showing that our method achieves optimality gaps of no more than 5% on these datasets. Moreover, the experimental results demonstrate that the quality of our solution surpasses that obtained through the ALNS method, and the disparity between them has been reduced by 2 to 43 times. This is because ALNS is a purely heuristic algorithm, whereas our method is a matheuristic approach.

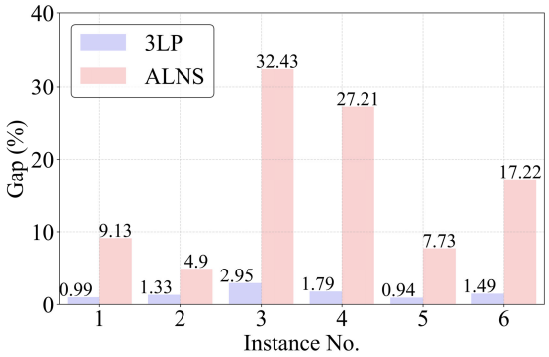
C. Computation Time

1) Algorithm Computation Time: Table III reveals that the computational speed of our proposed method significantly surpasses that of the ALNS method, achieving a speed improvement of two to three orders of magnitude compared to Gurobi. It is important to note, however, that the computation time of our method is influenced not only by the number of LP problems solved but also by the planning duration of the instances themselves. For example, the No. 8 instance, despite having the fewest LP problems to solve (only 6), required the longest computation time of 25.427 seconds. This is due to the planning duration of this instance being 60 hours, far exceeding the planning durations of the other instances in the Errico dataset. As the planning duration increases, the subproblem of inserting charging nodes requires more time to construct the model due to the need to consider a

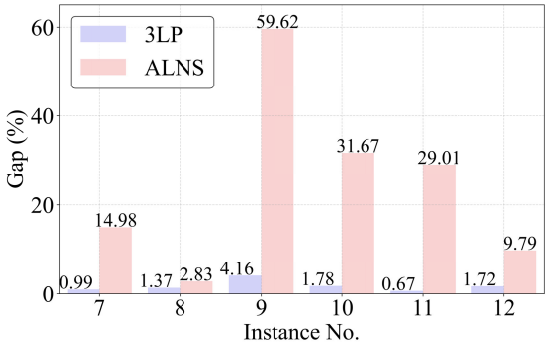
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TABLE III
EXPERIMENT RESULT FOR SMALL SCALE DATASET

No.	3LP					Gurobi				ALNS			
	t[s]	n_{LP}	Obj	TD[km]	CC[CNY]	t[s]	Obj	TD[km]	CC[CNY]	t[s]	Obj	TD[km]	CC[CNY]
1	4.265	12	-2086.634	859.402	54.564	7200	-2107.592	839.597	52.451	900	-1915.069	958.871	126.060
2	23.310	12	-2340.936	625.166	33.898	7200	-2372.513	600.245	27.242	900	-2256.369	705.804	37.827
3	12.621	21	-1068.557	1234.717	36.726	7200	-1101.053	1205.525	33.422	900	-744.020	988.686	37.294
4	37.266	15	-1391.819	1428.251	89.930	7200	-1417.192	1435.568	87.240	900	-1031.607	1713.959	254.434
5	3.449	25	-3192.409	2017.609	69.982	7200	-3222.553	1989.248	68.199	900	-2973.557	2093.003	93.440
6	6.331	17	-4120.824	1655.587	223.589	7200	-4183.113	1610.238	206.649	900	-3462.599	1967.858	329.542
7	5.301	12	-1860.779	1019.006	90.215	7200	-1879.317	1003.468	87.215	900	-1597.820	1203.982	198.198
8	25.427	6	-2320.919	644.730	34.351	7200	-2353.076	627.56	19.364	900	-2286.601	649.288	64.111
9	5.258	19	-1196.262	1311.660	42.078	7200	-1248.148	1316.284	45.568	900	-503.987	735.590	20.423
10	6.599	15	-1349.211	1419.549	141.240	7200	-1373.599	1397.476	138.925	900	-938.616	1722.084	309.300
11	3.471	19	-3913.246	1428.009	48.745	7200	-3939.464	1407.882	42.654	900	-2796.493	1883.798	79.709
12	4.535	14	-4579.816	1560.561	159.623	7200	-4660.034	1494.618	145.348	900	-4203.841	2266.88	459.279



(a) Solomon

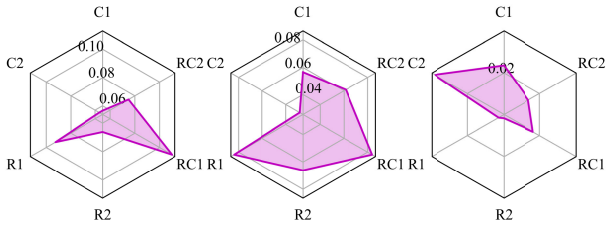


(b) Errico

Fig. 4. The optimality gap of EVRP-3LP algorithm and ALNS algorithm.

TABLE IV
EXPERIMENT RESULT FOR LARGE SCALE DATASET

No.	3LP				
	t[s]	n_{LP}	Obj	TD[km]	CC[CNY]
13	98.751	36	-149246.453	8128.844	224.703
14	168.790	48	-301010.127	11534.398	1055.475
15	165.119	19	-111840.978	7220.706	138.316
16	538.917	18	-142028.947	9303.84	283.213
17	155.864	30	-188415.172	10380.704	404.124
18	334.604	28	-216284.220	12334.015	597.765



(a) Solomon (b) Errico (c) Homberger

Fig. 5. The ratio of the actual number of LPs computed to the theoretical upper bound.

greater number of charging possibilities. This point is further validated by experiments on the Homberger dataset. As shown in Table IV, the No. 16 instance, despite having the fewest LP problems to solve (only 18), required the longest computation time (538.917 seconds). This is due to the planning duration for this instance being 7200 minutes. In summary, the experimental results demonstrate that our proposed method achieves exceptionally fast computational speeds when the planning duration is relatively short, fully meeting the requirements for online planning. For longer overall planning durations, we can further enhance computational efficiency by employing a rolling horizon decomposition approach, which will be a focus of our future research.

2) Number of LPs: Fig. 5 illustrates the ratio of the actual number of LPs computed to the theoretical upper bound of LPs. This ratio is defined by $\frac{n_{LP}}{n_{LP}^{UB}}$, where $n_{LP}^{UB} = 2|\mathcal{T}|+|\mathcal{K}|+2$ represents the theoretical upper bound. The experimental results indicate that n_{LP} is significantly smaller than n_{LP}^{UB}, demonstrating that the actual computational effort is substantially lower than the theoretical maximum.

D. Algorithm Reliability in Real Road Network

We validate the practicality of our proposed method and the reliability of the algorithm in real road network using the Belgium dataset. The Belgium datasets include two instances, i.e. the belgium-n50-k10.vrp and belgium-n1000-k20.vrp, which have 50 and 1000 customers, respectively. The experimental results are shown in Table V. These results indicate that our method demonstrates good adaptability to real applications.

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TABLE V
EXPERIMENT RESULT FOR REAL ROAD NETWORK DATASET

No.	3LP			
	t[s]	Obj	TD[km]	CC[CNY]
19	11.840	-1997.299	467.556	35.145
20	165.760	-23064.912	8354.464	480.624

V. CONCLUSION

In this study, we have investigated the Electric Vehicle Routing Problem (EVRP) with a specific focus on addressing the challenges associated with power capacity constraints in charging stations. Moreover, we recognize the frequent order changes in practical applications, underscoring the importance of computational efficiency in EVRP solutions. To meet these demands, we have introduced an innovative computational framework based on linear programming techniques, enabling rapid EVRP solutions. The iterative LP approach at the core of our algorithm ensures exceptional computational efficiency. The extensive simulations have provided compelling evidence of the efficacy of our proposed method. Notably, the results demonstrate the successful solution of large scale EVRP instances involving 800∼ 1000 customers in hundreds of seconds.

While the proposed approach effectively addresses key challenges in the EVRP, applying it to larger problem instances or real-time scenarios could present certain difficulties. A promising direction for future research is the integration of the rolling horizon method, which involves solving the problem over smaller time windows and using the results as warm starts for subsequent calculations, thereby improving scalability and adaptability in dynamic environments.

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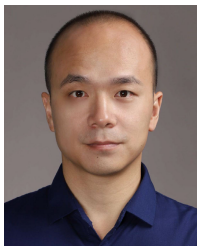
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